

Karthik Bangaru

B.Tech | Electronics & Engineering | JNTUA

✉ thekartheek@gmail.com | ☎ +91-77996 66552 | github.com/thekartheek | linkedin.com/in/thekartheek

EDUCATION

WorldQuant University (WQU), Washington D.C, USA

MSc in Financial Engineering | CGPA : 9.32 / 10

Jawaharlal Nehru Technological University (JNTUA) – KSRMCE

2021 – 2025

B.Tech in Electronics & Engineering | Applied Materials | CPI: 9.2 / 10

Swamy Trust , Puttaparthi

2019 – 2021

CBSE Class XII: 95.4% | Class X: 97.0%

SCHOLASTIC ACHIEVEMENTS

- National-level competitive **swimmer** (2023–2024), represented at state championship held in Mangalore
- Secured All India Rank 162 in JEE Advanced 2021 (out of ~1.4 lakh candidates)
- Awarded Institute Academic Prize for outstanding academic performance (Top 5% of batch)
- Selected for International Mathematical Olympiad (IMO) Training Camp, 2020

INTERNSHIPS & WORK EXPERIENCE

Indian Space Research Organisation (ISRO)

Satellite Imagery Testing | Intern

Nov 2025 – Apr 2026

Space Applications Centre | Tools: Python, OpenCV, GDAL, NumPy, QGIS, Matplotlib

- Developed automated validation pipelines for multispectral satellite imagery, reducing manual inspection time by 40%.
- Built preprocessing workflows for remote sensing data, including calibration, atmospheric correction, and cloud masking.
- Identified and documented 15+ critical defects in satellite image outputs, improving QA processes and mission readiness reviews.

Apple | Ads Developer | Remote

Feb 2024 – May 2025

Tools: Python, SQL, Apache Spark, Airflow, TypeScript, FastAPI

- Optimized large-scale ETL pipelines processing 500M+ daily ad events, reducing latency by 22%.
- Built an AI-driven QA tool for anomaly detection, improving data issue identification speed by 3×.
- Collaborated with ML teams on a Python-based bid calibration service and maintained 95% test coverage for reliable releases.

Software Engineering Intern – Google India, Hyderabad

May – Jul 2024

- Built a distributed caching layer for Google Maps serving 50M+ daily queries, reducing P99 latency by 35%
- Designed and implemented a real-time anomaly detection pipeline using Apache Kafka and PySpark
- Tech stack: Go, gRPC, Kubernetes, BigQuery, Prometheus

Research Intern – IIT Bombay (Prof. Sunita Rao, Systems Lab)

Dec 2023 – Jan 2024

- Implemented a custom memory allocator in C++ achieving 2× throughput over standard malloc on workloads
- Published findings as a workshop paper at SOSP-WiP 2024

KEY PROJECTS

SmartEdu – AI-Powered Personalized Learning Platform

Jan – Apr 2024

Course Project | Prof. Arjun Mehta | Tools: Python, React, FastAPI, PostgreSQL, OpenAI API

- Developed an adaptive quiz engine that adjusts difficulty in real-time using Item Response Theory
- Deployed on AWS (EC2 + RDS) handling 500+ concurrent users with 99.9% uptime during pilot

MiniOS – Bare-metal RISC-V Operating System

Aug – Nov 2023

Course Project | CS301 Operating Systems | Tools: C, Assembly, QEMU

- Implemented process scheduler (Round Robin + Priority), virtual memory (paging), and basic file system
- Passed 92/100 test cases on the autograder; ranked 3rd in class of 180 students

CryptoVault – Decentralized Password Manager

Self-initiated | 2023

Tools: Solidity, Ethereum, React, ethers.js

- Built a zero-knowledge proof-based password manager; keys never leave the client browser
- Open-source (200+ GitHub stars); presented at IIT Bombay Techfest 2023

TECHNICAL SKILLS

Languages: C/C++, Python, Go, Java, JavaScript/TypeScript, SQL, Bash

Frameworks & Libraries: React, FastAPI, Node.js, gRPC, Apache Kafka, PyTorch, scikit-learn

Tools & Platforms: Git, Docker, Kubernetes, AWS (EC2, S3, RDS, Lambda), GCP, Linux, PostgreSQL, Redis

Concepts: Distributed Systems, OS, DBMS, Computer Networks, ML/DL, Data Structures & Algorithms

POSITIONS OF RESPONSIBILITY

Convener, Web & Coding Club, IIT Kharagpur

Apr 2023 – Apr 2024

- Led a team of 25 members; organised 10+ workshops & 3 inter-college hackathons (1000+ participants)

Teaching Assistant – CS101 (Introduction to Programming)

Jul – Nov 2023

- Conducted weekly lab sessions for 60 students; evaluated assignments and held doubt-clearing sessions

EXTRACURRICULAR ACTIVITIES

- Competitive Programming: Codeforces rating 2100 (Expert); ranked in top 500 in CodeChef Long Challenge
- Participation in Smart India Hackathon 2023 – Finalist (national level, 10,000+ teams)
- Classical Guitar player; performed at IIT Madras Mood Indigo cultural festival (2022, 2023)